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Satvik M. Bhure

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Table of Contents:

Cover

Copyright

A Short Editorial

Question I: What is the future of Fusion Energy on Earth?

Question II: What is quantum physics and why is it so different to regular physics?

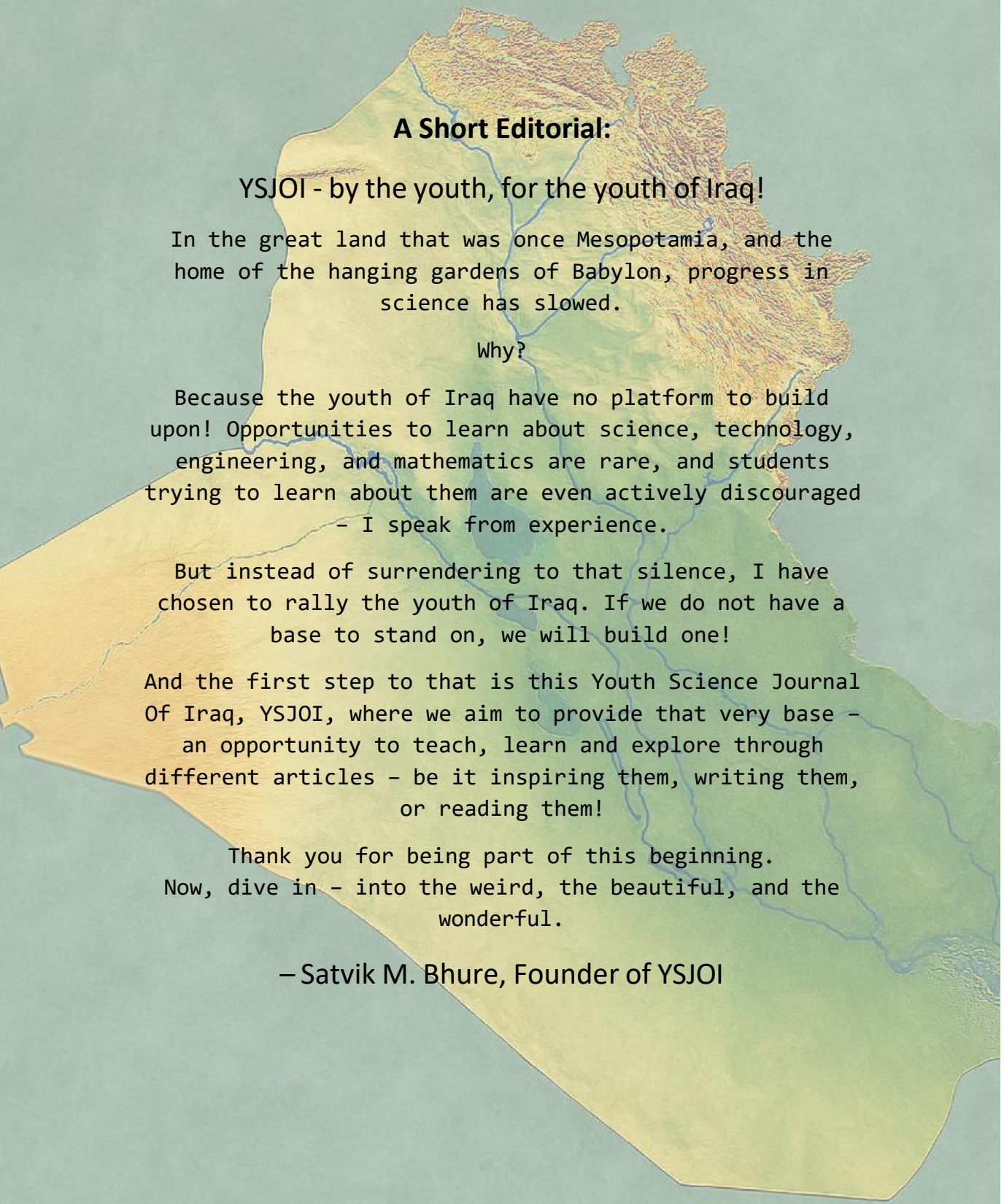
Question III: Who needs the other: The heart or the brain?

Question IV: What is the Dark Forest Hypothesis?

Question V: How do Dyson Spheres work?

Acknowledgements

Conclusion



A Short Editorial:

YSJOI - by the youth, for the youth of Iraq!

In the great land that was once Mesopotamia, and the home of the hanging gardens of Babylon, progress in science has slowed.

Why?

Because the youth of Iraq have no platform to build upon! Opportunities to learn about science, technology, engineering, and mathematics are rare, and students trying to learn about them are even actively discouraged – I speak from experience.

But instead of surrendering to that silence, I have chosen to rally the youth of Iraq. If we do not have a base to stand on, we will build one!

And the first step to that is this Youth Science Journal Of Iraq, YSJOI, where we aim to provide that very base – an opportunity to teach, learn and explore through different articles – be it inspiring them, writing them, or reading them!

Thank you for being part of this beginning.
Now, dive in – into the weird, the beautiful, and the wonderful.

– Satvik M. Bhure, Founder of YSJOI

Question I: What is the future of Fusion Energy on Earth?

– Muhammed Raad, 15

As the world faces climate change and increasing energy demand, scientists are searching for a cleaner and more reliable source of power. One of the most promising solutions is nuclear fusion. Fusion is the process that powers the Sun, where small atomic nuclei combine to form a heavier nucleus and release enormous amounts of energy.

Unlike nuclear fission, which splits heavy atoms and produces significant radioactive waste, fusion joins light atoms such as hydrogen isotopes. The reaction releases energy because a small amount of mass is converted into energy, as described by Einstein's equation, $E = mc^2$. This means fusion has the potential to produce far more energy than fossil fuels, without releasing carbon dioxide.

Fusion also has important safety advantages. It does not rely on a chain reaction, so it cannot melt down like a traditional nuclear reactor. In addition, its main fuel sources, such as deuterium, are widely available, especially in seawater. This makes fusion a potentially sustainable long-term energy solution for the planet.

However, achieving fusion on Earth is extremely challenging. The reaction requires temperatures of millions of degrees Celsius, creating a state of matter called plasma. This plasma must be controlled using powerful magnetic fields inside machines called tokamaks.

Maintaining stable plasma for a long time, while producing more energy than is used to heat it, is one of the biggest scientific and engineering challenges today.

Large international projects are currently working to make fusion energy possible. These projects aim to test whether fusion can produce net energy gain on a commercial scale. Although fusion power plants are not yet in operation, recent experiments have shown significant progress in plasma stability and energy output.

If these technological barriers can be overcome, fusion could transform the global energy system. It has the potential to provide clean, safe, and almost unlimited energy for future generations, reducing dependence on fossil fuels and helping to combat climate change.

Question II: What is quantum physics and why is it so different to regular physics?

– Wanya Ali, 12

Quantum physics is a science that studies the smallest things, even smaller than atoms. In the early 1900s when scientists like Albert Einstein and Max Planck realized that classic physics did not work at very small scales, the world of quantum physics was born – and it did not make *any* sense at first.

If you throw a ball, you can predict where it will go; but in the quantum world, particles do not behave as expected. They act like particles and waves at the same time. This idea is called wave particle duality. One of the most famous ideas in quantum physics is the uncertainty principle discovered by Werner Heisenberg. It says that you cannot know the exact position and the exact momentum of a particle at the same time. The more clearly you measure one the less clearly you know the other. It is not because our tools are bad, but because nature itself works this way, which can feel impossible to believe.

There is also something called superposition, which means a particle can exist in many possible states at once – until it is measured. A simple way to explain this is through the thought experiment by Erwin Schrödinger with his famous cat. The cat in the box with poison is both alive and dead until someone looks inside, because how could you ever prove otherwise?

Another strange effect is quantum entanglement, which happens when two particles become connected in such a deep way that measuring one instantly affects the other no matter how far apart they are. Albert Einstein once called this spooky action at a distance, because it seemed to break the rules of space and time.

Today, experiments have shown that entanglement is real and it is now being used in new technologies like quantum computing and secure communication – so that which was once impossible, becomes integrated in our daily lives!

So even though quantum physics might feel impossible to understand at first, even scientists say they do not fully understand it! So do not worry if you do not get it – after all, nobody really does.

Question III: Who needs the other: The heart or the brain?

– Ara Danyar, 17

Arguably, the two organs keeping you alive at this very moment are the heart and the brain. Sure, the others are vital – hence, the vitals – but none carry quite the same weight. So the question is irresistible: who matters more?

In one corner, the heart. The tireless muscle about the size of a small orange, beating roughly 100,000 times a day. It pumps liters of blood through a vast network of vessels, delivering oxygen to tissues, nutrients to cells, and carrying waste away to be removed. Every cell depends on it. Even the heart feeds itself through coronary arteries – vessels that take blood from the heart to the heart – because despite being only about 6 centimeters thick, it needs a constant supply to keep going.

The heart doesn't just move blood; it moves life. Hormones like insulin and glucagon travel through its currents to regulate blood sugar. Adrenaline, the spark behind fear and excitement, is rushed to where it's needed. Quietly, rhythmically, the heart sustains the entire system.

In the other corner, the brain. A soft, energy-hungry control center made of billions of neurons firing electrical signals. It governs your breathing, your blinking, your thoughts, and the simple act of lifting a cup of coffee. Different regions coordinate with astonishing efficiency, managing everything from memory to movement. And it does all this using only about 20 watts of power – less than a dim lightbulb.

So who depends on whom?

The heart has its own electrical circuit and can continue beating even without direct brain input. But the brain cannot survive long without oxygen-rich blood. Cut off its supply, and within about four minutes, irreversible damage begins.

Case closed? Not quite.

Without the brain, the body loses coordination, awareness, and control. Organs may function briefly, but the system collapses. A heart with nothing to support, nothing to sustain, beats in vain.

The truth is less dramatic but more profound: the heart and brain are interdependent. One supplies the power; the other directs it. Like most successful systems in nature, their strength lies not in competition, but in cooperation.

Question IV: What is the Dark Forest Hypothesis?

– Tiwan Ahmed, 12

Look up at the night sky. Billions of stars. Billions of planets. And statistically, at least a few of them should host life. So where is everyone? One possible answer is unsettling: they are out there. They're just quiet.

The Dark Forest Hypothesis imagines the universe as a vast, pitch-black forest. Every civilization is a hunter. Every hunter is armed. And no one knows who else is hiding between the trees. In such a forest, making noise is dangerous. If you shout to announce your presence, something stronger might hear you – and decide you are a threat. From this perspective, silence is survival.

Civilizations that broadcast their location might be wiped out before they can spread. The only ones that last are the cautious ones. The quiet ones. The ones who listen but do not speak. It would explain why the cosmos feels so empty. Not because it is lifeless, but because it is careful. It sounds dramatic, almost paranoid. But think about Earth's own history. When two civilizations met at vastly different technological levels, the outcome was rarely equal. Now stretch that imbalance across lightyears. Of course, this is only one idea among many. Maybe intelligent life is rare. Maybe it self-destructs. Maybe interstellar travel is too difficult. But the Dark Forest hypothesis forces us to ask an uncomfortable question: should we be shouting into space at all?

For now, the universe remains quiet. Whether that silence is peaceful – or watchful – we do not yet know.

And perhaps that is the most humbling thought of all: in a forest this large, even whispering might matter.

Of course, humans, being humans, are mostly rather ignoring this theory. We have yelled out into space, be it with radio waves or with our space probes – desperate, indeed, to find more travelers in this lonely night. And perhaps, indeed, we could set a trend; a trend of actually helping one another, and growing together with other intelligent life.

One day.

Question V: How do Dyson Spheres work?

– Rozho Michael, 16

Energy is life. Civilizations rise on it. Wood becomes coal. Coal becomes oil. Oil becomes nuclear fusion – eventually, perhaps. And if you follow that ladder far enough, you reach the ultimate power source: a star.

A Dyson Sphere is a theoretical megastructure built around a star to capture its energy output. Not necessarily a solid shell – that would collapse under its own gravity – but a vast swarm of satellites orbiting the star, collecting sunlight and converting it into usable power. Our Sun releases more energy in a single second than humanity has used in its entire history. Capturing even a fraction of it would make a civilization unimaginably powerful. Planets could be powered entirely by starlight. Industry, travel, computation – all scaled to levels we can barely picture.

In practice, building such a structure would require dismantling entire planets for raw materials. Countless solar collectors would orbit in coordinated patterns, beaming energy to habitats or storage systems. It would be engineering on a cosmic scale. And what would all that energy be used for? Exploration? Simulation? Interstellar travel?

Or, if we're being honest, probably just more advanced ways to boil water.

After all, from steam engines to nuclear reactors, much of our technological progress has involved finding creative methods to heat water and spin turbines. A Dyson Sphere might just be the universe's most ambitious kettle.

Still, the concept represents something bigger. It symbolizes a civilization that has moved beyond planetary limits. One that treats its star not as distant decoration, but as infrastructure.

If we ever see unusual dimming patterns from distant stars, some scientists speculate it could hint at such megastructures. A star partially veiled by its own technology.

Until then, it remains a powerful idea: that one day, intelligence might grow large enough to wrap itself around a sun – and call it progress.

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And we thank you, reader, for supporting our mission to help the Iraqi youth where there has been no help as yet!

CONCLUSION

We hope to further YSJOI's mission to empower the youth of Iraq in STEM fields by our monthly magazine, and sincerely hope we were able to help you and other students with this monthly issue! We are always open to feedback and questions, and everyone is of course welcome to learn in their own way; after all, we are all different, and we all belong.

Make sure to visit our website at https://YSJOI.is-great.org

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